NeuroBot: EEG-Based Communication Interface

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Abstract

Brain–computer interfaces (BCIs) offer a powerful path toward restoring communication for individuals with severe speech or motor impairments. In multilingual regions like India, however, most existing assistive technologies are built around English, limiting their accessibility for large native-language populations. This work introduces NeuroBot, an open-source pipeline designed to acquire, preprocess, and analyze EEG signals evoked by visually presented Hindi word stimuli. Using a 64-channel EEG system, validated artifact protocols, and accurate epochs, the pipeline provides a clean stimulus-locked dataset for offline classification based on classical machine-learning classifiers. The pipeline can reliably classify performance with low signal-to-noise ratios found in single-trial EEG by employing pseudo-trial techniques to boost/enhance discriminability. While a small number of deep learning strategies were compared to assess classification performance, traditional classifiers were significantly more consistently successful and informative for this high-dimensional, low sample dataset, particularly logistic regression. NeuroBot is a reproducible and customizable workflow that enhances the evidence base for inclusive, language-sensitive BCI studies. This work demonstrates feasibility and creates space for inclusive research and development of neural interfaces that are non-English, knowledge mobilization, and practical situating of BCIs within the communicative context of India.

Specifications table

Project name	NeuroBot
Field of study	Biomedical Engineering, Brain–Computer Interfaces, EEG Signal Processing, Machine Learning
Type of Software	Python-based EEG preprocessing and analysis pipeline for stimulus-locked decoding of visually presented Hindi words
Code repository	https://github.com/NeuroBot-DP
Programming language	Python (MNE-Python, NumPy, SciPy, scikit-learn, Matplotlib)
Supported file formats	BrainVision EEG format: `.vhdr, .vmrk, .eeg`

1 Motivation and Significance

Humans need to be able to communicate, but many people are unable to do so because of neurological conditions like stroke, paralysis, amyotrophic lateral sclerosis (ALS), or traumatic brain injury. According to the **Indian Council of Medical Research (ICMR)**, **7–12 million people** in India currently suffer from severe speech or communication impairments. Between **1.25 and 1.8 million** new stroke cases are reported each year, and about **60% of survivors have partial or complete speech and movement impairments**. These patients experience frustration, and a significant reduction in their quality of life because they are still cognitively aware but **unable to communicate verbally**.

By using **mechanical or sensor-driven solutions**, such as **eye-tracking systems**, **switch-operated devices**, or **voice synthesisers**, current assistive technologies aim to lessen this communication barrier. Despite being moderately successful, these methods usually have a number of drawbacks, such as relying on residual motor function, requiring a great deal of training, or being too expensive for general use in settings with limited resources. Furthermore, in **multilingual regions like India, accessibility and adaptability are limited** because most of these technologies are designed for English or Western contexts.

Brain-Computer Interfaces (BCIs) provide a new way for users to communicate by examining brain activity to determine user intent. BCIs heavily based on electroencephalography (EEG) are advantageous in their increased temporal resolution, portability, affordability, and non-invasive technique. EEG records changes in electrical potential due to brain functioning which can then be evaluated to understand motor or cognitive targets. Nevertheless, and despite advancements in research, there are still notable obstacles in bringing EEG based BCIs from experimental simulations to reliable approaches for communication in the real world. These challenges include signal noise, variability amongst subjects, a complexity of the configuration, and the absence of a shared open source experience.

At this level of evolution, NeuroBot targets the acquisition of EEG data and offline analysis. With the 64 channel EEG cap, we acquired stimulus locked EEG data and created a standardized preprocessing pipeline composed of filtering, re-referencing, artifact handling, and epoching. The datasets of cleaned and labelled epochs were utilized to train and evaluate an offline machine learning model with the intention of determining whether distinct patterns of EEG behaviour could be established with the visual stimuli presented to the participant. In all, it establishes a standardized and reproducible approach for data acquisition, preprocessing, and model training, which is the first step towards more advanced BCI capabilities in subsequent work.

In conclusion, NeuroBot takes a innovative step towards how brain computer interfaces are imagined in India. **Instead of relying on the usual English centric datasets** that dominate global BCI research, this project deliberately **works with Hindi words**, acknowledging the linguistic reality of millions of Indian users. Very few attempts have been made to explore how the human brain responds to **region specific languages in a BCI setting**, and NeuroBot directly addresses this gap. By combining **standardized EEG acquisition** with an openly accessible processing pipeline, the system proves that neurotechnology does not have to speak only one language. It can adapt to the people who use it. NeuroBot doesn't just decode signals it opens the door for multilingual, culturally rooted neurotechnology, something that has been long overdue in India's research landscape. This work establishes a foundation for future systems that think beyond conventions, embrace Indian languages, and move toward truly inclusive BCI innovation.

2 Software Description

The software developed for this project provides a clean and dependable workflow for processing EEG data collected during a controlled word presentation experiment. At this stage, the focus is deliberately narrow: import the raw EEG recordings, clean them, extract well aligned epochs, and train a simple offline model to check if the data contain learnable structure. While the long term goal is a more capable decoding pipeline, this version concentrates on building a stable and transparent foundation that future work can grow from. The entire pipeline is implemented in Python using MNE-Python, which offers a robust and research-validated environment for working with EEG signals.

2.1 Software Architecture

The EEG dataset was recorded using a 64-channel ANT Neuro, eego64 system following the international 10–10 electrode layout. A total of 10 participants took part in the experiment. All recordings used CPz as the acquisition reference, an EOG channel to track blinks, and a 1000 Hz sampling rate to preserve high-resolution temporal detail. Electrode impedances were maintained below 20 kΩ throughout each session.

Participants were shown Hindi words presented visually on a screen, with random digits inserted occasionally to ensure attention. After each block, participants used a simple slider interface to report how many digits they had noticed. Each participant completed 3000 trials, from a vocabulary of 100 unique Hindi words. The ANT system automatically embedded event markers for every stimulus presentation, allowing precise synchronization between stimuli and EEG activity.

Recordings were stored in ANT's BrainVision format (.vhdr, .vmrk, .eeg), which the software can open directly with no conversion.

2.2 Software Functionalities

The software pipeline is organized into four major stages:

1. Data Loading and Inspection
2. Preprocessing of the Continuous EEG
3. Epoching and Labelling
4. Offline Model Training and Evaluation
5. Evoked Response (ERP) Analysis
6. Topographic Maps (100–300 ms)

This modular structure keeps the workflow simple to follow while leaving plenty of room for future extensions such as feature extraction or real-time decoding.

2.2.1 Data Loading and Inspection

The very first step in the pipeline is Data Loading and Inspection. All that the software does is load the raw EEG recordings directly from ANT's BrainVision format (.vhdr, .vmrk, .eeg), so there will be no additional conversion of the data.

Once the data has been loaded, the software provides you with the necessary tools to check the quality of your data, which include:

1. **Time Series Plot:** The software provides the raw EEG signal for multiple channels (i.e. EEG Fp1-CPz, EEG Fp2-CPz, etc.) as a function of time in one of the figures (top image on page 5), where you can see both the continuous signal and visible stimulus event markers (triggers on bottom) that are shown on the plot.

2. **Power Spectral Density (PSD) Plot:** The PSD Plot is the bottom of page 5. The plot shows the power of the signal (in dB/Hz) and the frequency (in Hz - x-axis). The step of the inspection is very important because it shows the noise profile of the data immediately after it occurs.

Time series plot with triggers on bottom

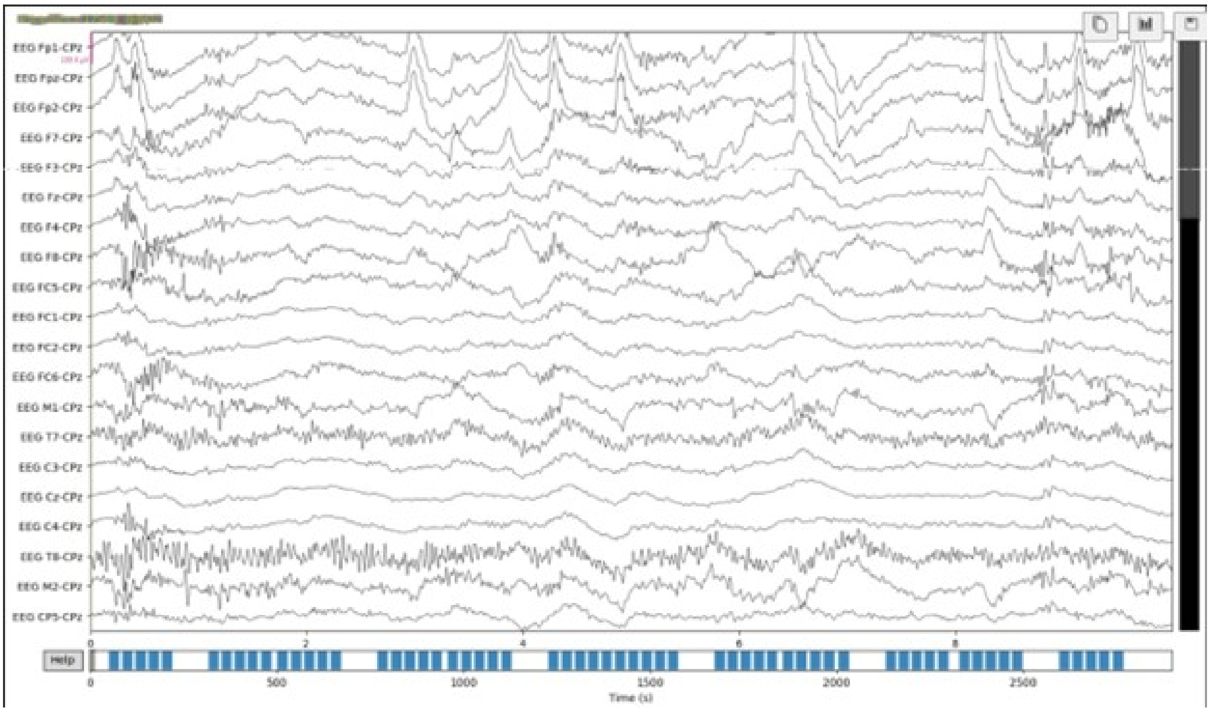


Figure 1: EEG Electrodes Signals

PSD plot

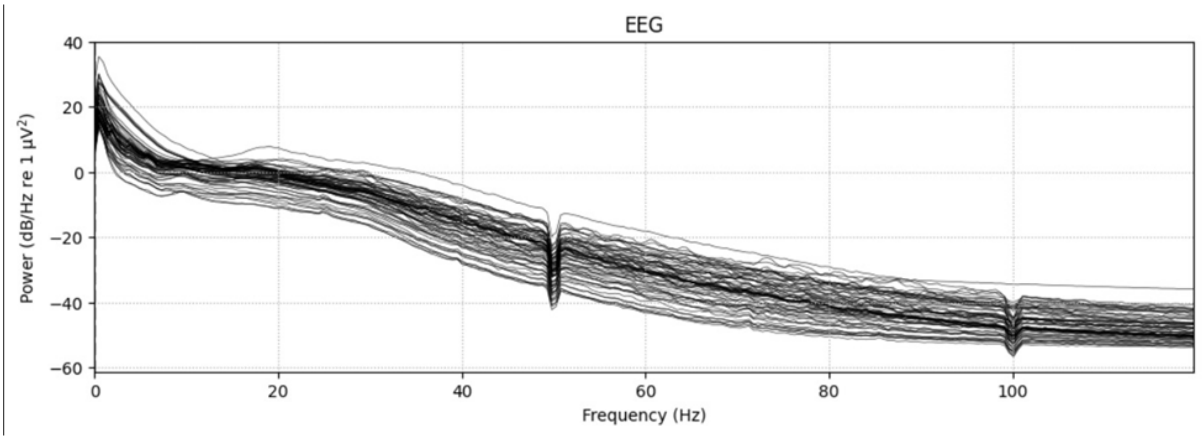


Figure 2: PSD Plot

PSD plot after notch filtering 50 Hz notch

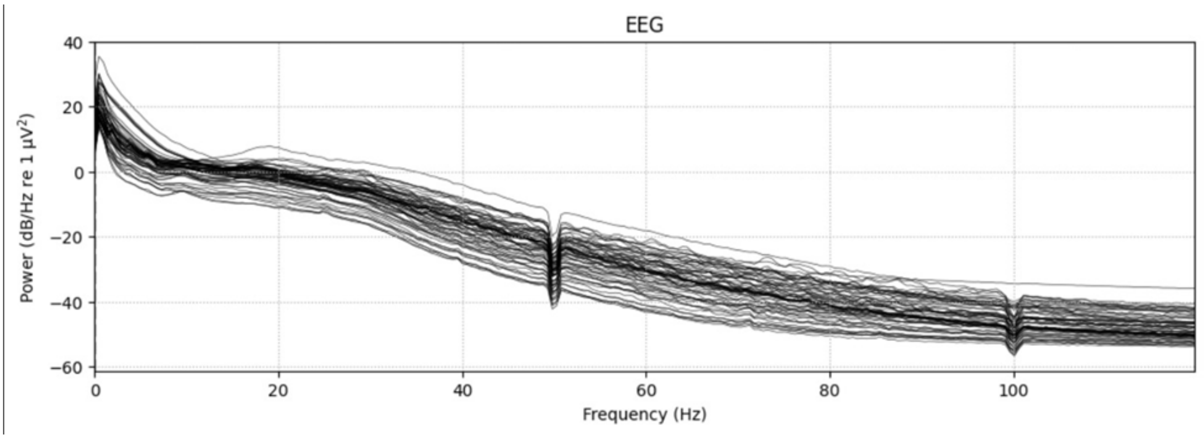


Figure 3: PSD plot after notch filtering 50 Hz notch

PSD plot after bandpass 0.3 to 20 Hz

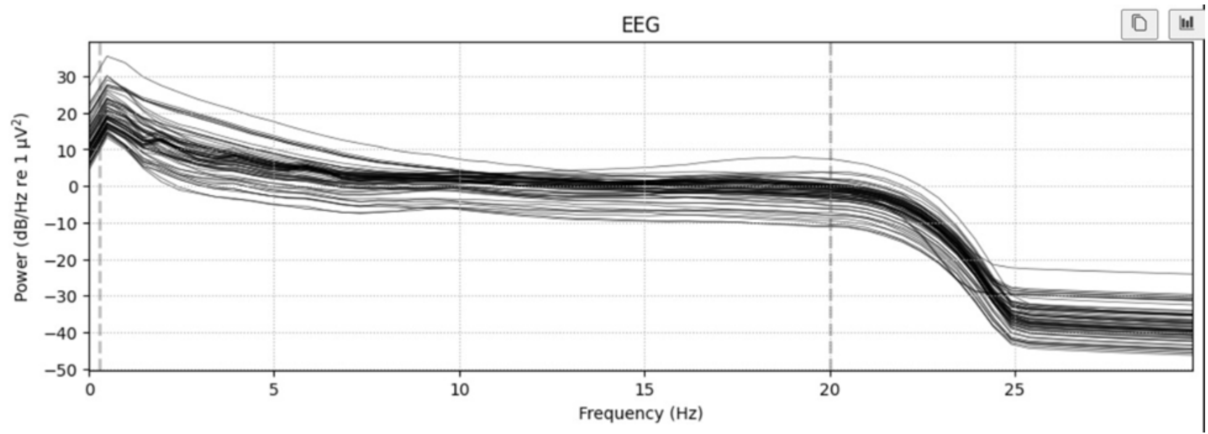


Figure 4: PSD plot after bandpass 0.3 to 20 Hz

Topomaps for 4 Random Epoch

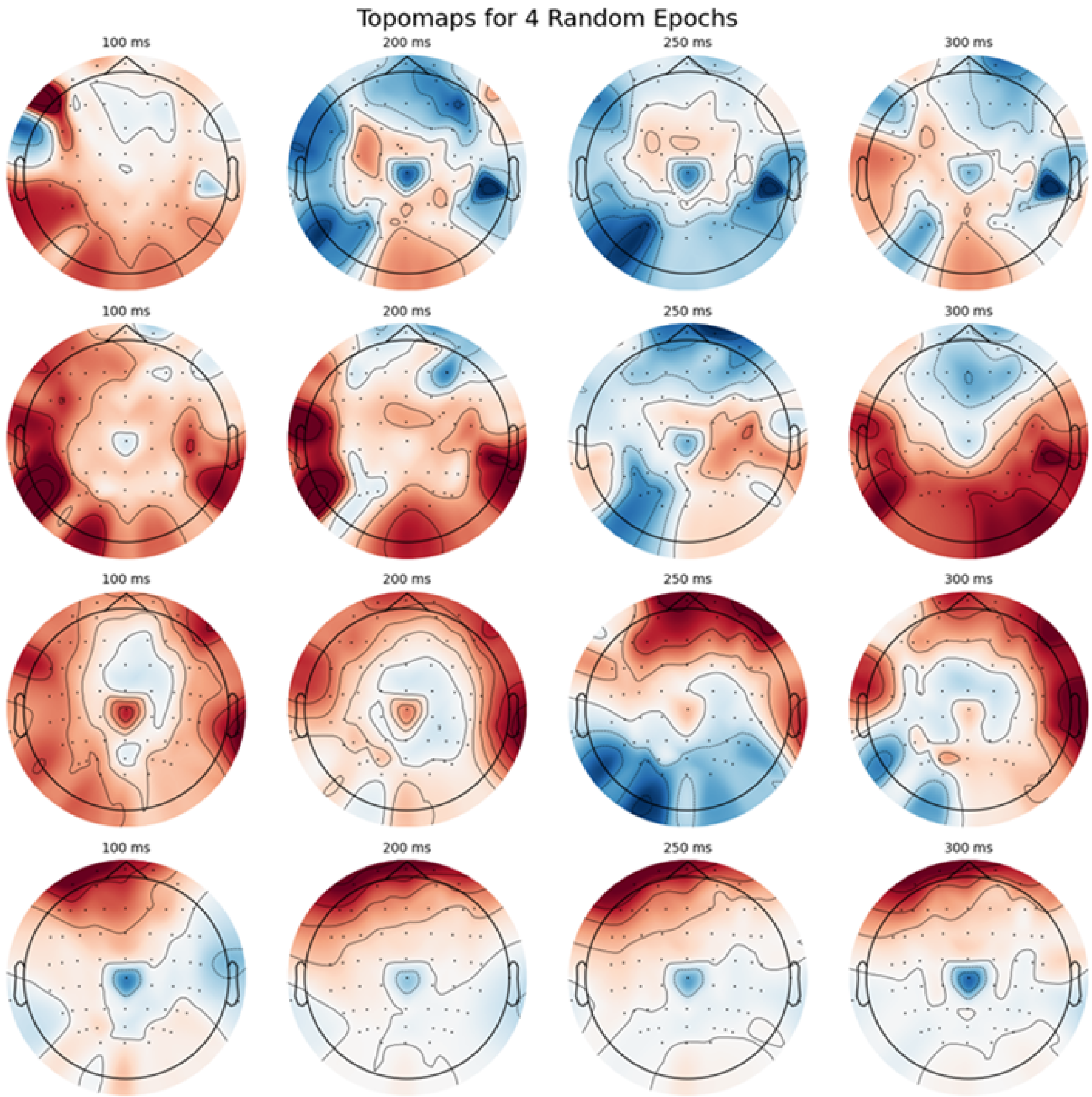


Figure 5: **Topographical Maps for four Epochs**

2.2.2 Preprocessing of the Continuous EEG

The following preprocessing steps were applied to the raw EEG data:

1. 50 Hz notch filter
2. Bandpass filter (0.3–20 Hz)
3. Epoching from -100 ms to 400 ms
4. Baseline correction using the -100 to 0 ms interval
5. Label assignment using synchronized log files

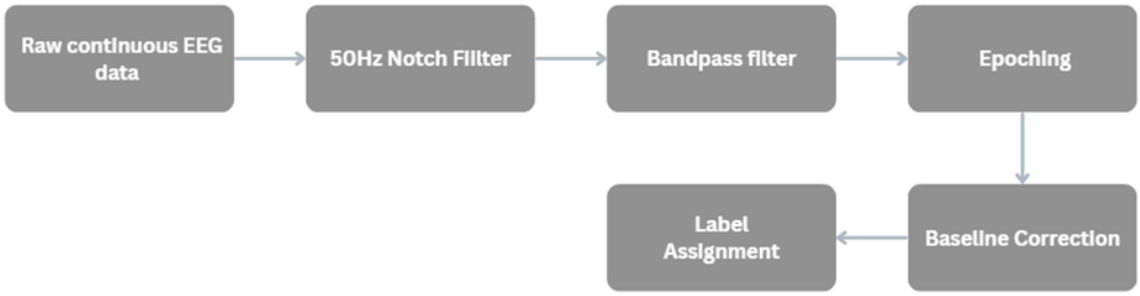


Figure 6: EEG Acquisition Pipeline

2.2.3 Epoching and Labelling

We first performed epoching around each stimulus presentation to create meaningful training examples for our EEG decoding task. Each of the stimulus words was presented for a fixed amount of time on the screen during the experiment. Later, we used the log file generated from PsychoPy to extract the exact timestamps of those stimulus onsets. Using these timestamps, we aligned each EEG segment (epoch) with the correct label corresponding to the word that was displayed during that particular time window.

The following table is one example of the stimulus-level metadata, used during the labelling process. The columns represent the properties of each stimulus that were mapped to the respective EEG epochs: Word, Frequency_Class, MN Label (1/0), and Length_Class.

	A	B	C	D
1	Word	Frequency_Class	mn label(1,0)	ength_class
2	पुलिस	1	1	1
3	जिला	1	1	0
4	थाना	1	1	0
5	सड़क	1	1	0
6	विद्यालय	1	1	1
7	बिजली	1	1	1
8	कार्यालय	1	1	1
9	कार्यक्रम	1	1	1
10	भवन	1	1	0
11	दुकान	1	1	1
12	पंचायत	1	1	1
13	प्रेमी	1	1	1
14	अपराधी	1	1	1
15	शोभायात्रा	1	1	1
16	मुहल्ला	1	1	1
17	राजमार्ग	1	1	1
18	विशेषज्ञ	1	1	1
19	पुरस्कृत	1	1	1
20	आश्रम	1	1	0

Figure 7: Dataset for Stimulus

2.2.4 Offline Model Training and Evaluation

1. Machine Learning Pipeline Due to the low signal-to-noise ratio in single-trial EEG, we adopted a pseudo-trial (Pseudotrial, sometimes also mentioned as SuperTrials in this paper) strategy to improve classification performance. Pseudotrials were created by averaging random samples of trials from the same class to improve the Signal to Noise Ratio and bring out the structure in the Signal. We took care to not use the same set to create the Training and testing Pseudo Trials to avoid any kind of data leakage. Logistic Regression, SVM, and ensemble models were tested, with Logistic Regression performing best.

We evaluated three decoding settings:

1. Pseudotrial → Pseudotrial (upper bound)

2. Pseudotrial → Raw (realistic)
3. Raw → Raw (single-trial baseline)

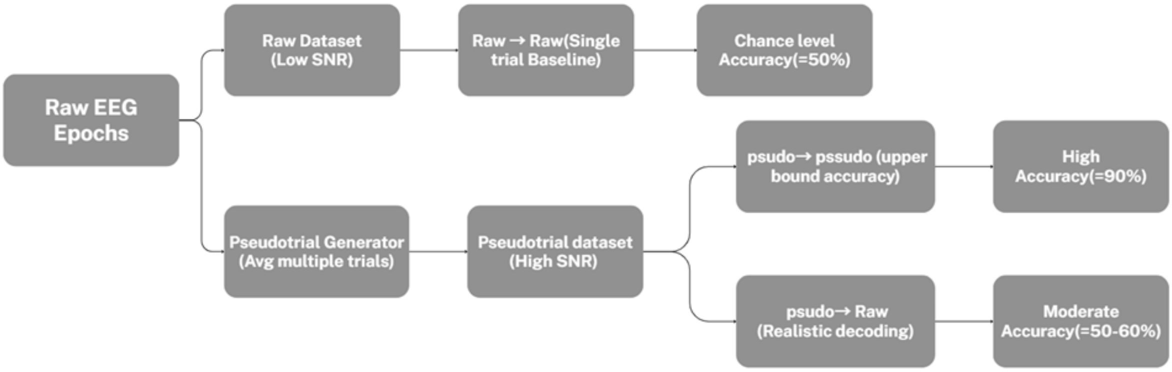


Figure 8: Machine Learning Pipeline

2. Key Results

- **Pseudo → Pseudo:** Accuracy ≈ 90%, AUC ≈ 98%

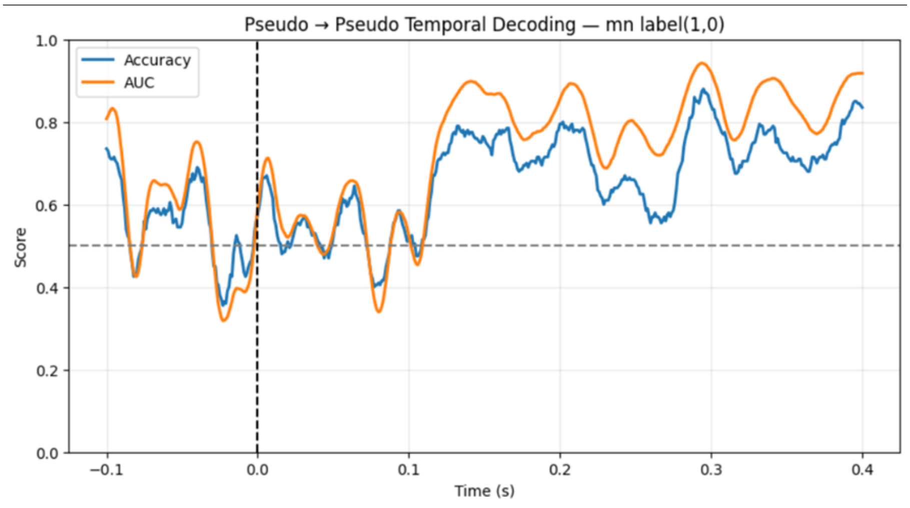


Figure 9: Pseudo to Pseudo results

- **Super → Raw:** Accuracy ≈ 60% (Max), 53% (Average)
- **Raw → Raw:** Performance at chance level (≈ 52%)

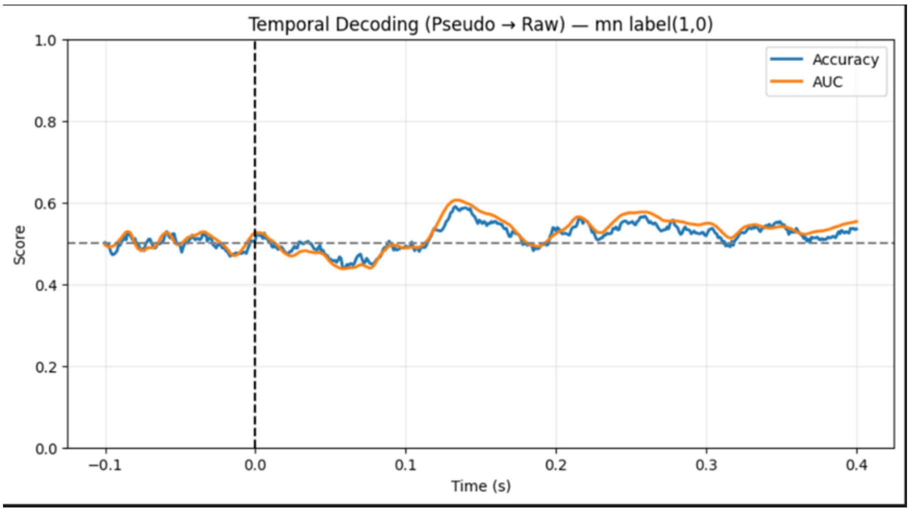


Figure 10: Pseudo to Raw results

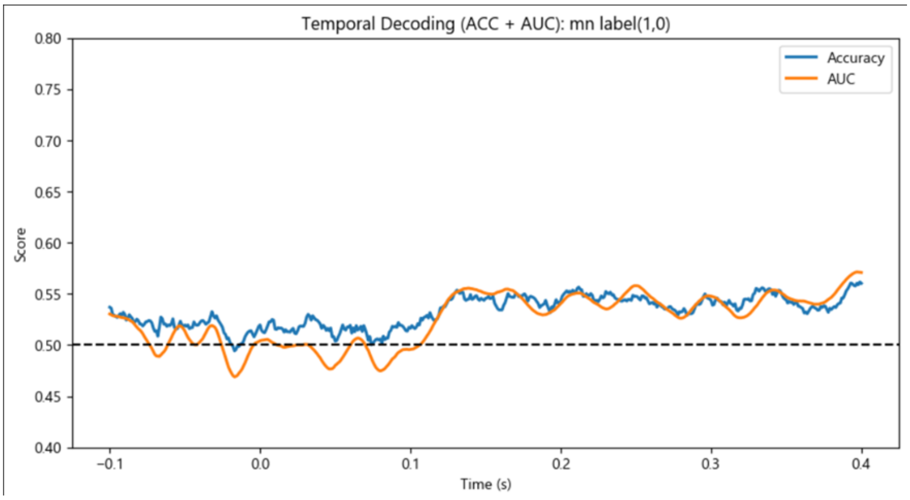


Figure 11: Raw to Raw Results

These results confirm that the brain reliably encodes lexical category information, but single-trial decoding remains difficult due to low SNR.

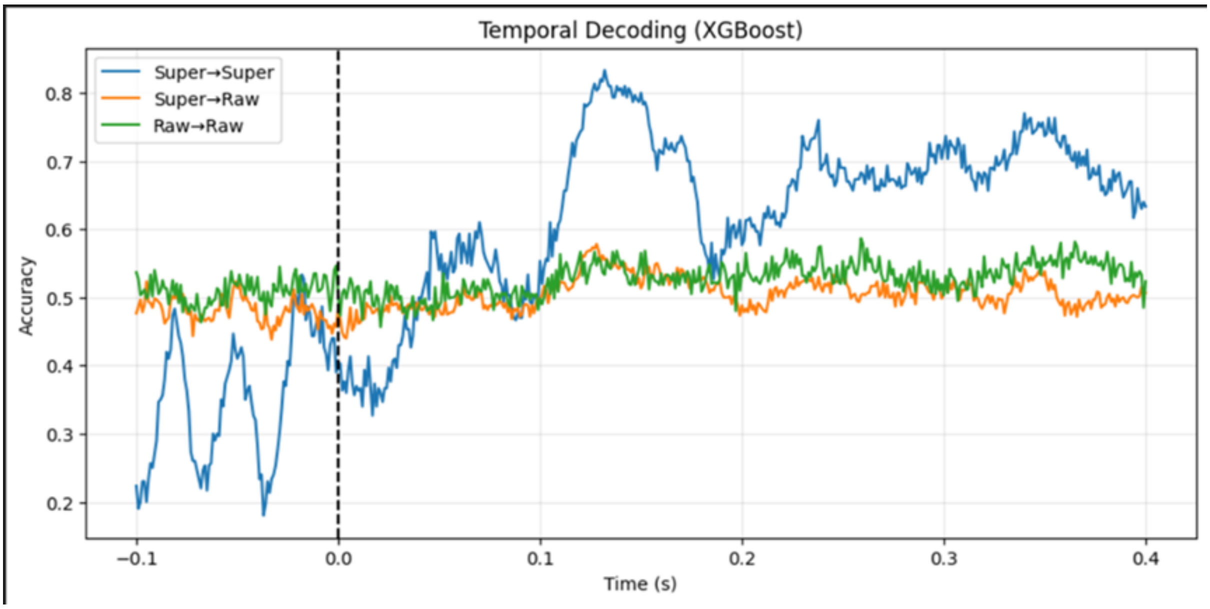


Figure 12: Temporal Decoding using XGBoost

3. Confusion Matrices and Classification Reports **Confusion Matrix: Frequency_Class Decoding (Super→Raw @ t=0.128s)**

Confusion Matrix: Frequency_Class Decoding (Raw→Raw @ t=0.259s)

------ Super→Super @ 0.132s (acc=0.833) ------

	precision	recall	f1-score	support
0	0.8521	0.8067	0.8288	150
1	0.8165	0.8600	0.8377	150
accuracy			0.8333	300
macro avg	0.8343	0.8333	0.8332	300
weighted avg	0.8343	0.8333	0.8332	300

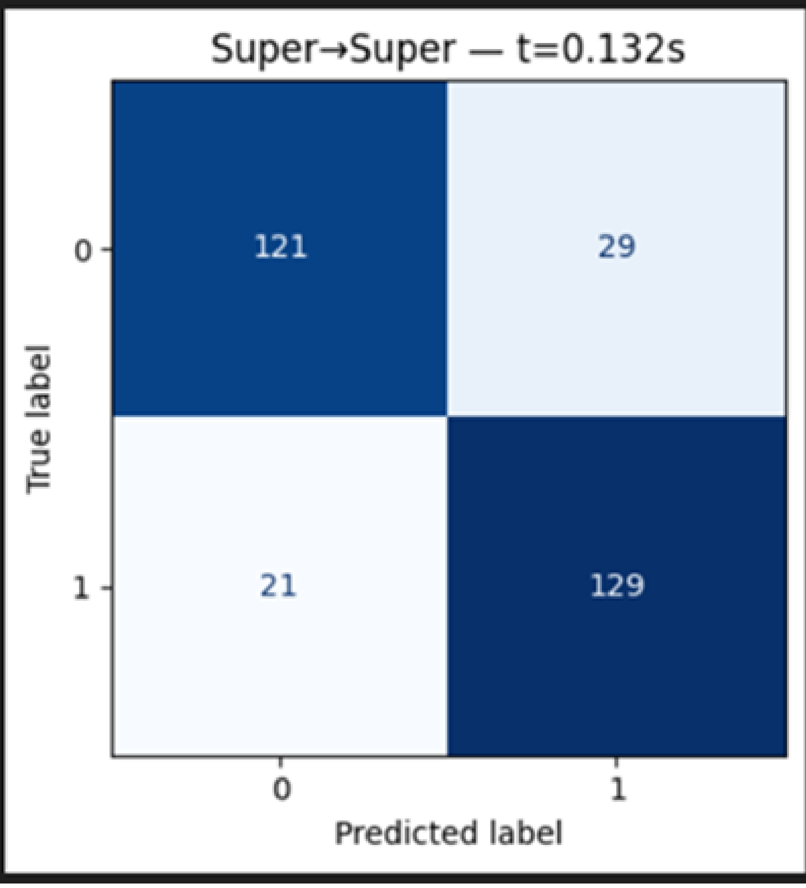


Figure 13: Confusion Matrix: Frequency_Class Decoding (Super→Raw @ t=0.128s)

------ Raw→Raw @ 0.259s (acc=0.587) ------

	precision	recall	f1-score	support
0	0.5524	0.4296	0.4833	270
1	0.6051	0.7152	0.6556	330
accuracy			0.5867	600
macro avg	0.5788	0.5724	0.5694	600
weighted avg	0.5814	0.5867	0.5781	600

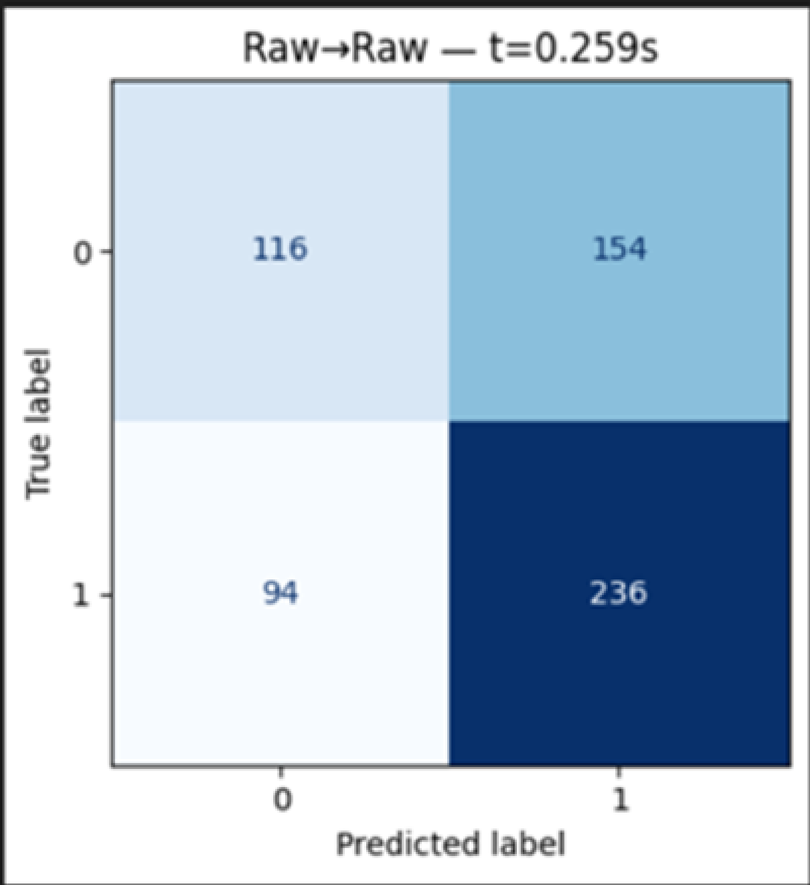
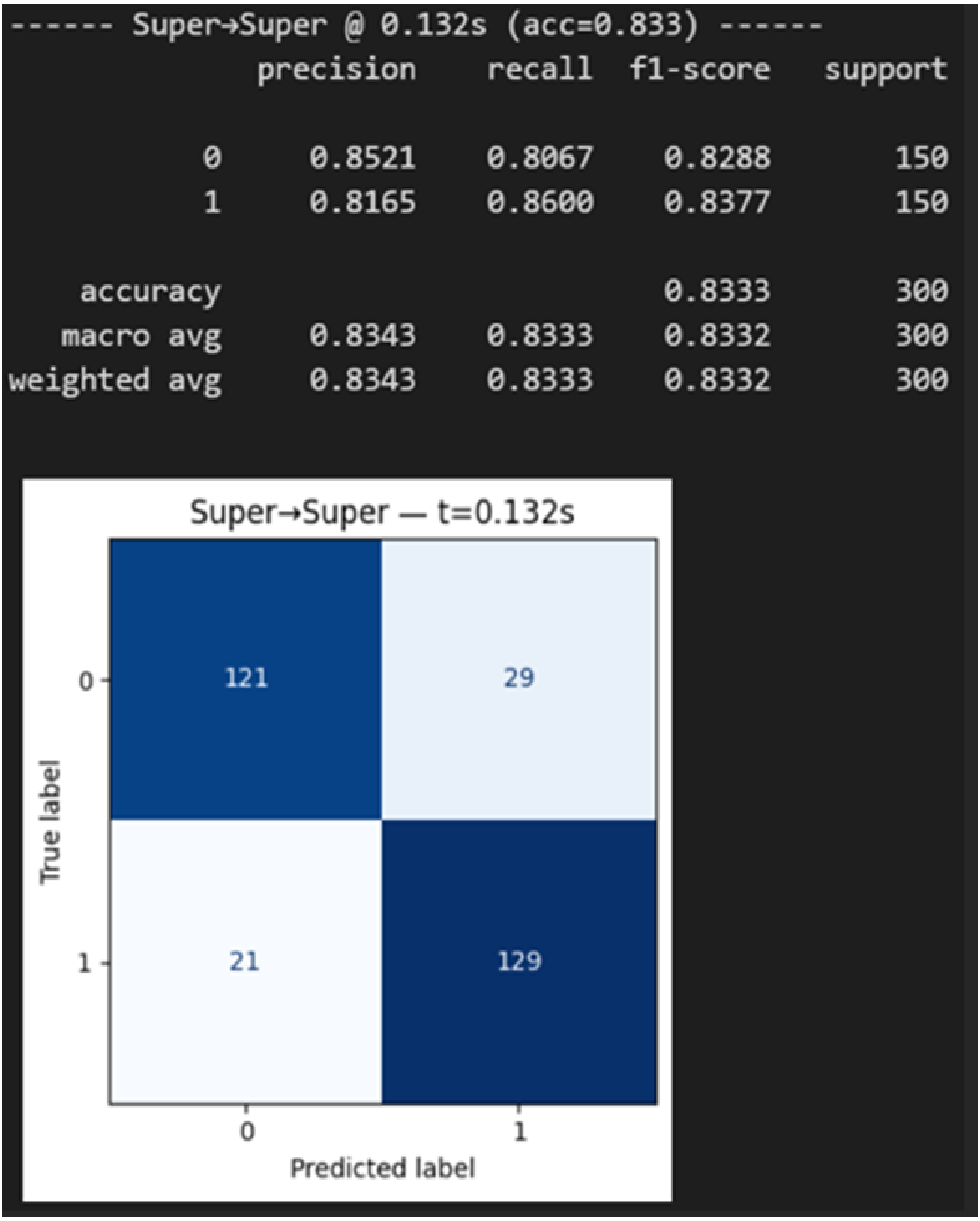


Figure 14: Confusion Matrix: Frequency_Class Decoding (Raw→Raw @ t=0.259s)

Confusion Matrix: Frequency_Class Decoding (Super→Super @ t=0.132s)



------ Super→Super @ 0.132s (acc=0.833) ------

	precision	recall	f1-score	support
0	0.8521	0.8067	0.8288	150
1	0.8165	0.8600	0.8377	150
accuracy			0.8333	300
macro avg	0.8343	0.8333	0.8332	300
weighted avg	0.8343	0.8333	0.8332	300

Figure 15: Confusion Matrix: Frequency_Class Decoding (Super→Super @ t=0.132s)

4. Signal-to-Noise Ratio (SNR) Observations Pseudo-trials significantly improve SNR ($\propto \sqrt{n}$). Raw trials have very low SNR and are insufficient for reliable word-level decoding. Future work will focus on advanced denoising, such as xDAWN, ICA, and Riemannian geometry methods.

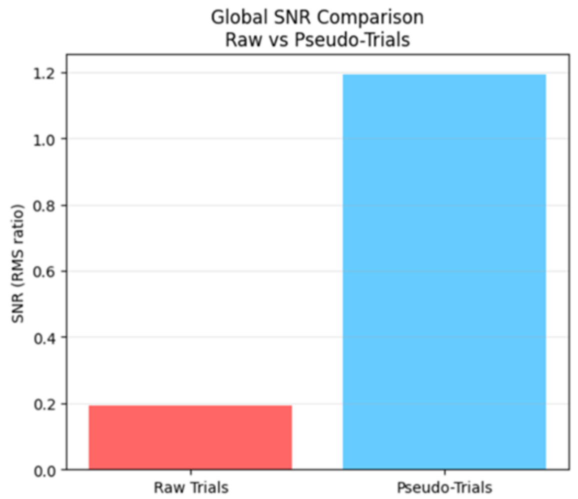


Figure 16: SNR Comparison

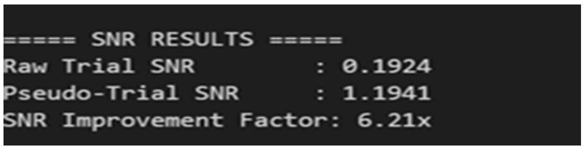


Figure 17: SNR Results

2.3 Deep Learning Attempts (EEGNet, CNNs, LSTM, Transfer Learning)

We also experimented with multiple deep learning architectures to test whether they could outperform classical machine-learning models. The following models were implemented and evaluated:

- EEGNet
- ShallowConvNet
- 1D CNN + LSTM hybrid network
- Fully Connected Neural Network (MLP)
- Transfer Learning using ResNet-50 (on time–frequency images)

However, all deep learning models suffered from one or more of the following issues:

1. Serious Overfitting Due to the small dataset size and high dimensional input, CNNs and MLPs quickly memorized the training pseudo-trials. Even regularization with dropout, weight decay and batch norm did not solve the training curves.

- High training accuracy (>95%)
- Very poor test accuracy (<55%)

2. Poor Generalisation to Raw Trials Models trained on pseudo-trials were unable to decode raw single-trial EEG, mainly because:

- Pseudotrials have high SNR
- Raw trials have very low SNR
- Deep networks expect stable distributions, which EEG does not provide

3. High Computational Latency Some architectures (LSTM, CNN-LSTM, ResNet50) introduced:

- High inference time per trial
- Large preprocessing overhead (e.g., STFT or spectrogram generation)

This directly conflicts with the requirement of real-time BCI decoding, where latency must remain below ∼100–200 ms.

4. Models That Didn't Overfit Underperformed EEGNet and ShallowConvNet were more stable, but their decoding accuracy on raw trials remained poor (<55%), and training them on 3000 trials took significantly longer than classical ML models.

Conclusion From Deep Learning Experiments Deep learning models did not outperform classical machine-learning approaches such as Logistic Regression, SVM, or XGBoost on this dataset.

Given:

- Small dataset size
- Low SNR on each trial
- High chance representation variability across words
- Need to perform in real-time

We decided to use classical ML models that resulted in:

- Faster training
- Lower latency
- Better generalization
- More stable performance
- More interpretability

Subsequent NeuroBot versions may revisit deep learning:

- After larger multi-subject datasets were collected
- After applying more advanced spatial filtering (i.e. xDAWN, CSP, Riemannian Geometry)
- Or using self-supervised representation learning

2.4 Evoked Response (ERP) Analysis

We computed evoked responses (ERPs) for each of the three binary classes. Clear differences in neural activity emerged around 250 ms, matching the latency of cognitive ERP components like the N200 and N400 reported in literature.

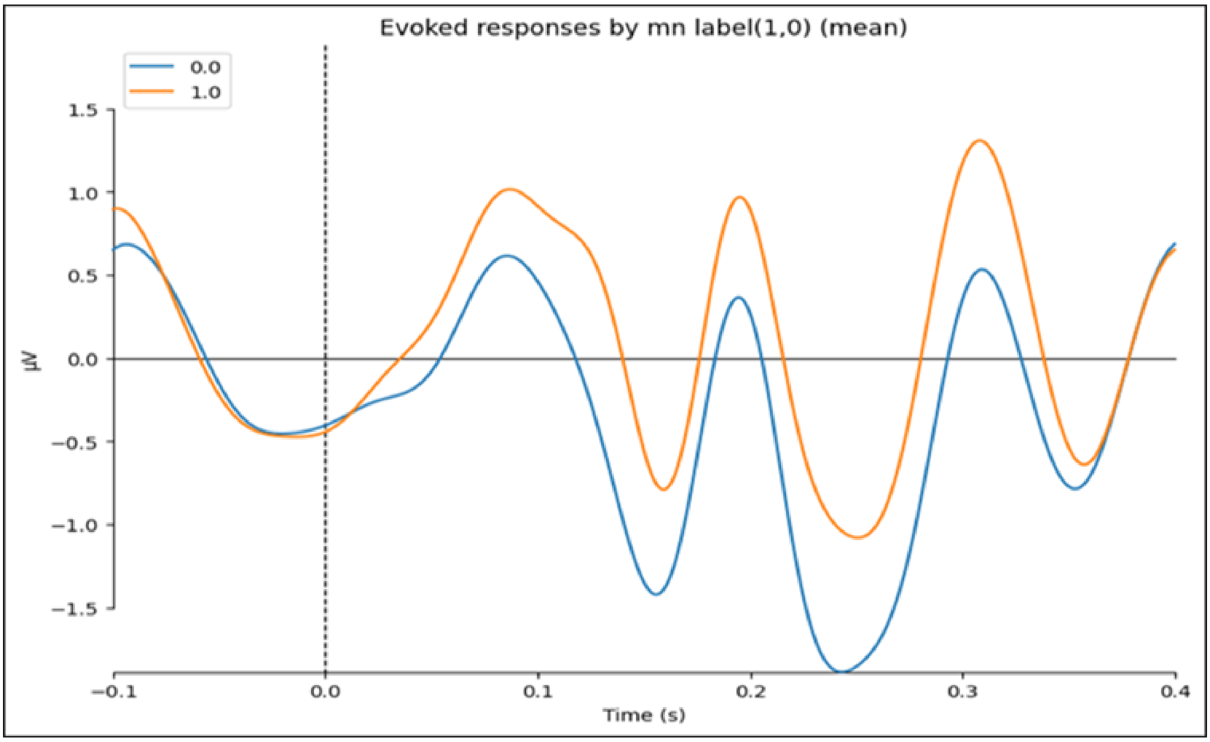


Figure 18: ERP (man-made labels)

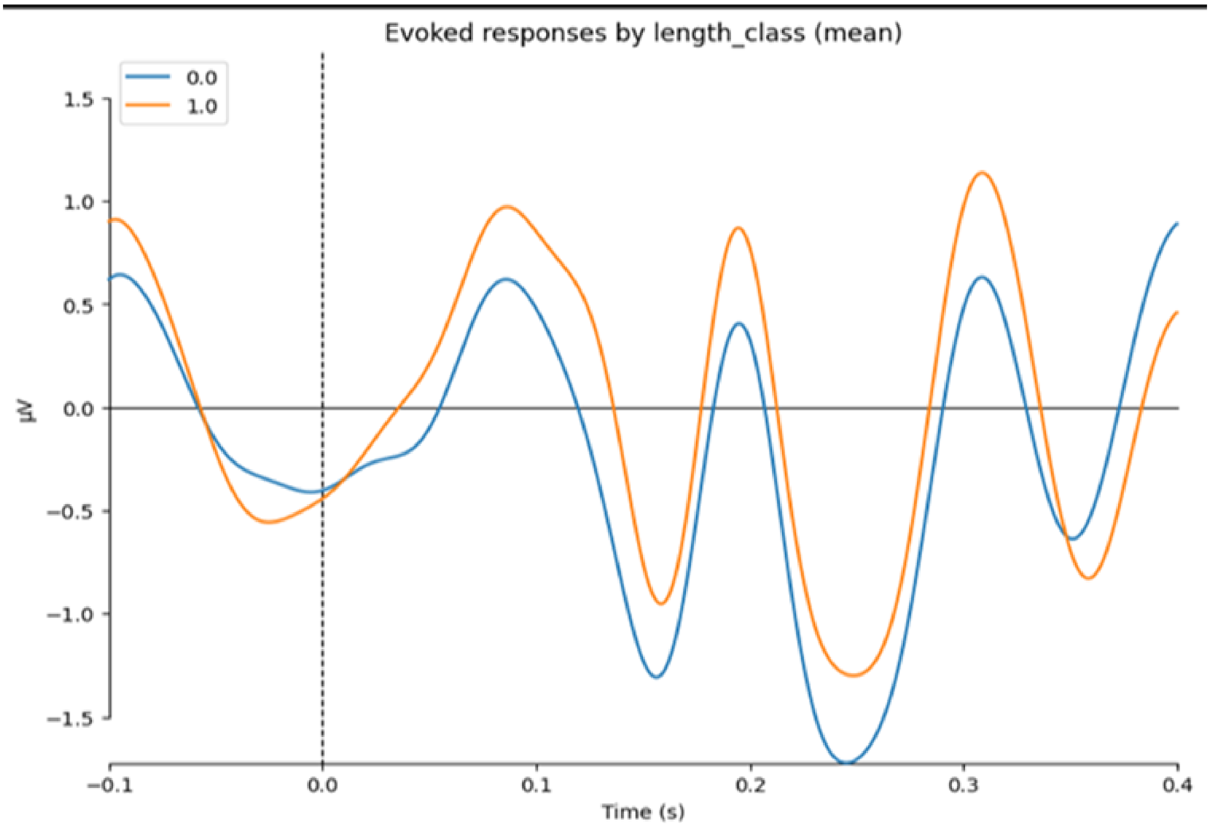


Figure 19: ERP (Word Length)

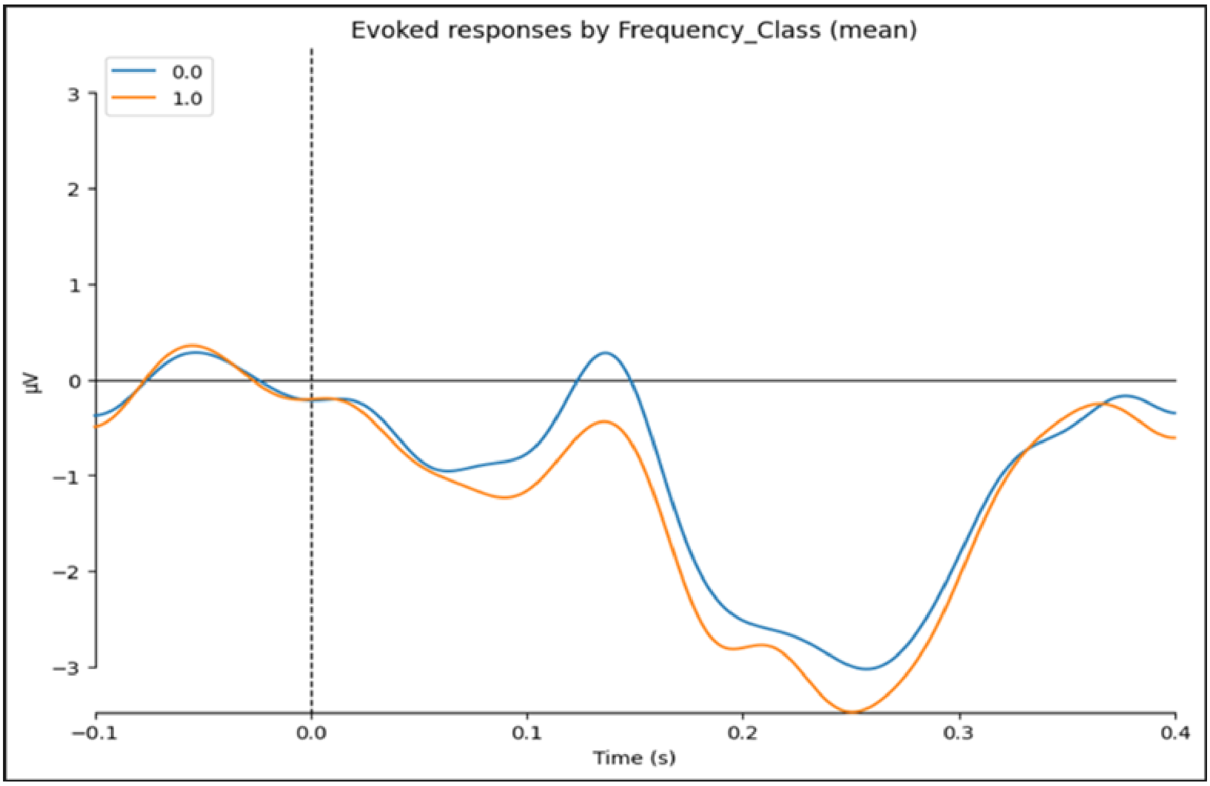


Figure 20: ERP (Lexical Frequency)

3 Future Scope

Our future goals include:

- Improving single-trial decoding using advanced preprocessing
- Exploring deep learning models (EEGNet, ShallowConvNet)
- Cross-subject generalization
- Real-time decoding pipeline development
- Integration with robotic control modules

Future research in the NeuroBot pipeline will pursue several directions to bring a viable treatment, from an analysis tool, to a functional, real-world system. The primary goal will be on improving single-trial decoding, which will be done with sophisticated preprocessing and denoising techniques, as this was determined to be a real limitation due to low Signal-to-Noise Ratios (SNR) in raw trials. As noted earlier, goals were set on implementing advanced preprocessing and denoising techniques, like xDAWN, ICA and Riemannian metric based methods, for each participant and trial.

At the same time, work will be a re-exploration of deep learning models including robust architectures like EEGNet and ShallowConvNet. This approach will be explored again upon collecting larger, multi-subject datasets and improved input data quality achieved through the advanced spatial filtering applied.

Additional future goals from the original proposal include verifying cross-subject generalization to build models that would apply for new users without significant retraining and a real-time decoding pipeline, which is beyond the current offline analyses and is an important step for making BCI functional. The main goal from the original proposal was to apply our findings through a final integration with robotic control modules, ultimately developing a complete, inclusive communication pathway.

4 Impact

This work makes a meaningful contribution to EEG-based communication research in India. By building a structured dataset using visually presented Hindi words, we take one of the first concrete steps toward understanding language-specific EEG responses outside the usual English-dominant BCI space. This helps address a clear gap in the field and brings much-needed diversity to EEG research.

The software created in this project converts raw EEG recordings into clear, organized, and usable data through a simple and reliable process. Instead of depending on specialized proprietary tools, researchers can load, clean, re-reference, epoch, and analyze recordings using one open-source pipeline. This helps students, new researchers, and labs with limited resources work confidently with EEG data.

One of the key outcomes of this work is showing that meaningful and distinguishable patterns can be extracted from EEG responses to Hindi word stimuli. This demonstrates that high-quality, language-relevant EEG research is entirely achievable with accessible hardware and a simple experimental design. It opens the door to future efforts in multilingual BCIs and assistive communication technologies designed for Indian users.

Although this version focuses on offline analysis, the workflow forms a strong base for what comes next—whether that's feature extraction, deep learning models, or early real-time decoding experiments. Overall, the work shows that linguistically grounded EEG research can be carried out reliably and that the groundwork for more advanced decoding tools is now firmly in place.

5 Conclusions

This work presents **NeuroBot**, an open-source EEG framework built around Hindi language stimuli—an area almost entirely overlooked in current BCI research. By focusing on India's linguistic landscape, NeuroBot addresses a clear gap in a field dominated by English-only datasets and demonstrates that culturally specific BCIs are both necessary and achievable. Using standardized EEG acquisition, a clean preprocessing pipeline, and offline classification, we show that neural responses to Hindi words contain consistent and learnable structure, even with the expected challenges of low single-trial SNR.

Within this dataset, traditional machine-learning methods, such as logistic regression and SVM, were more reliably successful than deep learning approaches, which reflects the realities of high-dimensional EEG and small data sets. Adding pseudo-trial averaging led to a notable improvement in accuracy, illustrating a pragmatic process for early-stage decoding. These findings provide a realistic reliable baseline for future work toward more complex or real-time systems.

Beyond its contributions in technical terms, NeuroBot advances the notion of inclusive, locally relevant neurotechnology. For millions of individuals in India with speech and movement impairments, the development of accessible BCIs must start from tools written in their own languages and context. NeuroBot is the first step in that process.

Future efforts will expand toward improved single-trial decoding, better cross-subject generalization, deeper learning architectures, and eventual integration with assistive communication or robotic systems. Overall, this work reinforces that building **multilingual, culturally aligned BCIs** is both feasible and essential, and it sets the stage for broader global progress in equitable brain-computer interface development.

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